

Missing Person Detection from Low-Resolution Drone Images using Super-Resolution and Detection

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Abstract

This paper investigates the effect of super-resolution (SR) on human detection in low-resolution drone imagery using various detection-classification pipelines. We evaluate four configurations: (1) Only YOLO, (2) $SR \rightarrow YOLO$, (3) $YOLO \rightarrow SR \rightarrow ResNet Ver.1$ (trained on YOLO-generated labels), and (4) $YOLO \rightarrow SR \rightarrow ResNet Ver.2$ (trained on ground-truth labels). Applying SR to the full image did not improve detection performance and led to high computational cost. Ver.1 suffered from label noise and domain mismatch, resulting in very low precision and mAP. In contrast, Ver.2 significantly improved both metrics by using high-quality labels and aligning training and inference domains. These results suggest that the effectiveness of SR depends heavily on its integration strategy and training data quality.

1. Introduction

Detecting missing persons from aerial drone footage is a critical yet challenging task due to low resolution, small object sizes, and visual similarity with the background. These factors hinder the ability of even state-of-the-art detectors like YOLO to accurately localize and classify targets.

To address these challenges, we explore the use of image super-resolution (SR) techniques within the detection pipeline. While SR improves visual clarity for human observers, its effect on machine-based detectors is not well understood. Thus, we evaluate several pipeline architectures integrating SR at different stages.

We experiment with four configurations: (1) baseline YOLO, (2) applying SR to the full image before YOLO ($SR \rightarrow YOLO$), (3) applying SR after YOLO-detection and classifying each region using a ResNet trained on YOLO predictions (Ver.1), and (4) a similar setup using ground-truth labels to train the ResNet classifier (Ver.2). Our results highlight the critical role of label quality and domain alignment in achieving effective SR integration.

2. Methodology

2.1. Dataset and Preprocessing

We utilized the VisDrone dataset, a large-scale aerial imagery dataset collected from drones over urban and natural scenes. It includes 6,471 training images, 548 validation images, and 805 test images. For simplicity, we grouped the 10 original object classes into two categories: *person* and *non-person*. Images were resized and normalized to meet the input format of each model. For ResNet training, cropped bounding box images were created using either YOLO predictions (Ver.1) or ground-truth annotations (Ver.2).

2.2. Pipeline 1: Only YOLO

In the baseline pipeline, we trained a YOLO model on the VisDrone dataset without applying any super-resolution. The model was modified for binary classification and trained using standard YOLO losses: objectness, bounding box regression, and classification. Inference was performed directly on the original low-resolution images.

Qualitative Results of Only YOLO

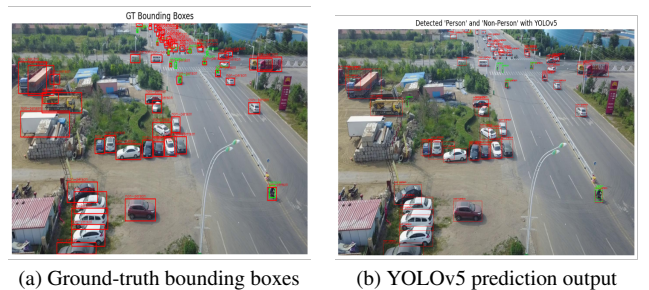


Figure 1. Visual comparison between ground-truth labels and YOLOv5 predictions. Some person instances are missed or misclassified.

2.3. Pipeline 2: SR $\rightarrow$ YOLO

Here, we applied Real-ESRGAN, a state-of-the-art image super-resolution model, to the entire input image before passing it to YOLO. The intention was to enhance low-resolution details that may help detection. However, due to domain mismatch and excessive inference time (about 12s per image), this approach yielded inferior performance compared to the baseline.

Qualitative Results of Only SR $\rightarrow$ YOLO

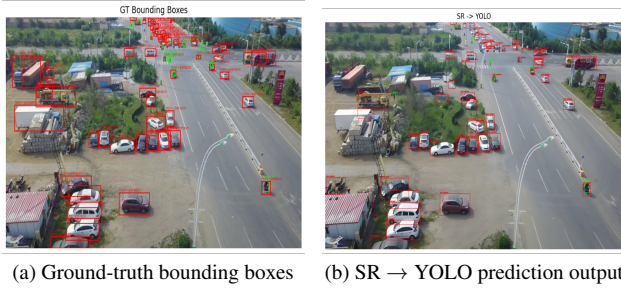


Figure 2. Comparison between ground-truth and YOLO predictions after applying SR.

2.4. Pipeline 3: YOLO $\rightarrow$ SR $\rightarrow$ ResNet Ver.1

In this pipeline, we first used YOLO to detect bounding boxes from low-resolution input. Each detected region was cropped and passed through Real-ESRGAN to produce a super-resolved patch. A ResNet-18 classifier was then used to classify each patch as person or non-person. For Ver.1, the training dataset for ResNet was constructed using YOLO’s own predictions as labels. As a result, the classifier inherited many labeling errors. Moreover, it was trained on low-resolution patches while inference was done on SR-enhanced inputs, causing a domain mismatch.

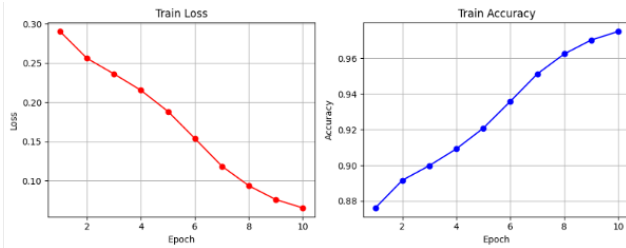


Figure 3. Training loss and accuracy curves of the ResNet classifier in Pipeline 3. The model showed clean convergence during training, indicating that it effectively learned the provided data. However, poor pseudo-labels and a mismatch between training and inference inputs led to degraded test performance.

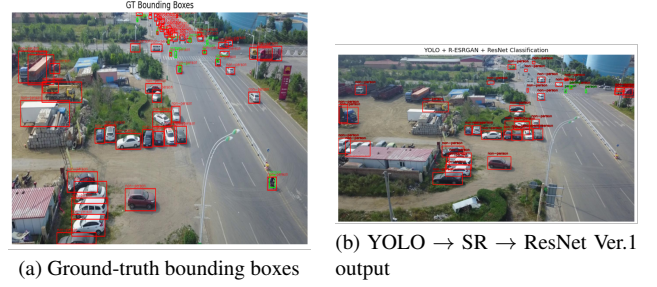


Figure 4. Comparison between ground-truth and final predictions from Pipeline 3 (ResNet Ver.1). Misclassification and missed detections are frequent.

2.5. Pipeline 4: YOLO $\rightarrow$ SR $\rightarrow$ ResNet Ver.2

To address the limitations of Ver.1, we constructed the ResNet training dataset using ground-truth bounding boxes and labels. We also trained the ResNet classifier using SR-enhanced patches to match the inference condition. With high-quality labels and domain alignment, Ver.2 achieved significant improvements in precision (from 20.22% to 48.39%) and mAP (from 0.1766 to 0.7753). This demonstrates the critical role of label accuracy and consistent data preprocessing.

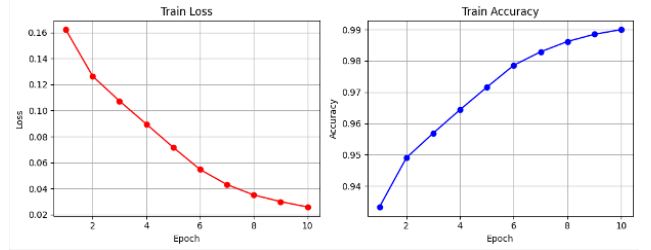


Figure 5. Training curves of ResNet classifier in Pipeline 4 (Ver.2). With accurate ground-truth labels and SR-enhanced inputs, the model converged quickly and reached high accuracy.

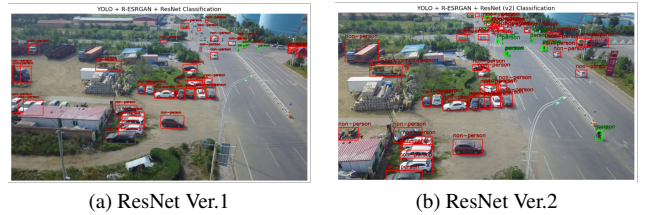


Figure 6. Visual comparison between Ver.1 and Ver.2. With cleaner labels and SR-matched training, Ver.2 shows improved classification and reduced false positives.



Figure 7. Qualitative comparison of all pipelines. The Ver.2 pipeline (YOLO $\rightarrow$ SR $\rightarrow$ ResNet) detects persons missed by other models (green circles), showing improvement potential. However, misclassifications still occur when persons overlap with similar-looking objects (blue circles).

Figure 7 illustrates a qualitative comparison across models. The Ver.2 pipeline successfully detects certain persons that were missed by the other approaches, especially in challenging areas like sidewalks or occluded zones. This indicates the potential for performance improvement using SR-assisted classification. However, the model still struggles to correctly classify persons that are visually overlapped or surrounded by similar objects (e.g., signs, poles), suggesting that further refinement in classification robustness is needed.

3. Experiments and Results

3.1. Evaluation Metrics

We evaluated model performance using the following metrics:

- **Precision:** The proportion of positive predictions that were correct. High precision indicates fewer false positives.
- **Recall:** The proportion of actual positive instances that were correctly detected. High recall indicates fewer false negatives.
- **F1 Score:** The harmonic mean of precision and recall, balancing the two.
- **mAP@0.5:** The mean Average Precision at IoU threshold 0.5, measuring localization and classification quality.
- **Inference Time:** Average processing time per image.

3.2. Quantitative Results

Table 1 summarizes the performance of the four pipelines.

Among the four evaluated pipelines, the Only YOLO configuration demonstrated the most balanced performance in terms of both accuracy and efficiency. It achieved the highest recall and F1 score, as well as the fastest inference time (50ms), making it ideal for real-time scenarios.

In contrast, applying super-resolution to the entire image (SR $\rightarrow$ YOLO) significantly increased inference time to over 12 seconds, while offering no gain in precision or recall due to domain mismatch.

The Ver.1 pipeline, which applies SR after YOLO and uses pseudo-labels for ResNet training, resulted in the lowest precision and mAP. This degradation stems from noisy labels and input inconsistencies between training and inference.

Ver.2 mitigated these issues by training ResNet with accurate ground-truth labels and SR-enhanced inputs, resulting in precision of 48.39% and mAP of 0.7753. Although its inference time (240ms) is higher than the baseline, it remains feasible for practical use.

Pipeline	Precision	Recall	F1	mAP@0.5	Time (ms)
Only YOLO	49.37%	23.69%	0.3202	0.8322	55
SR $\rightarrow$ YOLO	41.56%	23.62%	0.3012	0.7833	12390
YOLO $\rightarrow$ SR $\rightarrow$ ResNet (Ver.1)	20.22%	19.34%	0.1977	0.1766	3627
YOLO $\rightarrow$ SR $\rightarrow$ ResNet (Ver.2)	48.39%	22.20%	0.3044	0.7753	4210

Table 1. Quantitative performance comparison of each pipeline.

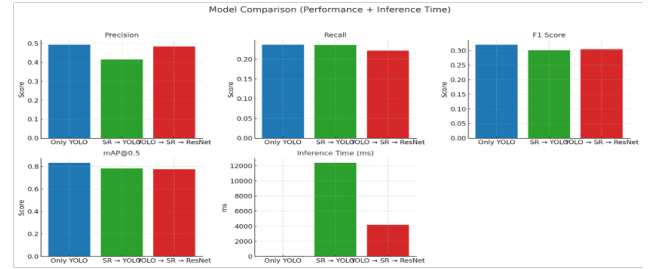


Figure 8. Overall comparison of all four pipelines in terms of accuracy metrics and inference time. Only YOLO performs best overall, while YOLO $\rightarrow$ SR $\rightarrow$ ResNet (Ver.2) achieves strong precision and mAP with moderate speed.

3.3. Qualitative Analysis

In visual comparisons, the Ver.2 pipeline detected persons that were missed by other models, particularly in cases where objects were small and blended with the background. This suggests that high-quality SR inputs combined with accurate training labels allow the classifier to focus on subtle human features.

3.4. Discussion

Our experiments indicate that naive integration of super-resolution may not improve—and can even harm—detection performance. SR before detection suffers from domain shift and computational inefficiency. In contrast, SR applied after bounding box localization, when combined with accurate labels and domain-consistent training, can enhance classification performance. This highlights the importance of pipeline design and label reliability when leveraging SR techniques in object detection tasks.

4. Conclusion

In this work, we investigated the impact of super-resolution (SR) on human detection from low-resolution drone imagery. Our experiments show that naive integration of SR—particularly when applied to full images—can degrade performance and introduce latency. While SR improves perceptual quality, it may disrupt feature distributions expected by detection models.

However, when SR is applied after object localization and combined with high-quality ground-truth labels (as in Ver.2), we observed a substantial improvement in classification performance. This suggests that SR can be effective if properly integrated within the detection pipeline, with consideration for training conditions and label quality.

Our findings emphasize the importance of pipeline design, domain alignment, and label reliability in SR-assisted detection systems.

YOLO-style detectors extract local gradient features such as $\partial I/\partial x$ and $\partial I/\partial y$ in their early convolutional layers. These gradients form the basis of hierarchical feature maps, which are used to build shape priors for bounding box regression. The accuracy of detection therefore heavily relies on the integrity of edge information within the input image.

However, when super-resolution is applied before detection, it may inadvertently distort these edge structures. SR models like Real-ESRGAN are optimized for perceptual quality—enhancing visual clarity for human viewers by improving color contrast and reducing noise (see Figure 9)—but at the cost of suppressing edge sharpness and blurring fine object boundaries (Figure 10). As a result, the shape priors learned during training may no longer align with the input features, leading to unstable bounding box predictions and degraded mAP.

To investigate this, we compare edge maps between original and super-resolved images. This offers a task-specific and intuitive way to assess how SR alters gradient consistency, which cannot be captured through RGB-level visual inspection alone.



Figure 9. Comparison between original (left) and SR-enhanced (right) image. SR improves perceptual clarity and color fidelity, benefiting human perception.

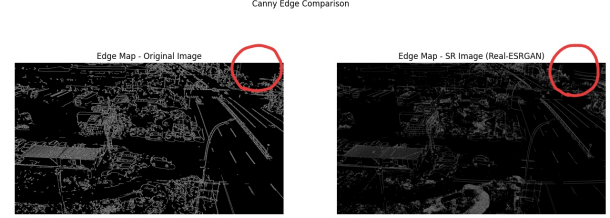


Figure 10. Edge map comparison of original (left) and SR-enhanced (right) images. SR slightly blurs object boundaries, reducing structural clarity that detectors depend on.

Fortunately, in the YOLO $\rightarrow$ SR $\rightarrow$ ResNet (Ver.2) pipeline, SR is applied after object localization. This architecture avoids disrupting YOLO’s performance and instead leverages SR for classification, where visual fidelity helps. Because the classifier is trained on SR-enhanced inputs, it is better adapted to handle the altered feature distribution.

These findings suggest that SR has potential in detection systems if integrated strategically and modularly. Future work may explore stronger classifier architectures and task-specific SR techniques designed for machine consumption rather than human aesthetics.

Moreover, as illustrated by the red circle in Figure 10, certain contours that were not detectable in the original image became visible after applying super-resolution (SR). This suggests that while SR may slightly attenuate strong edges, it tends to enhance weaker ones. Such enhancement is likely a key factor enabling Ver.2 to detect persons that other models failed to identify.

Future work may explore stronger classification architectures such as ResNet-50 or Vision Transformers, as well as end-to-end training schemes tailored for SR-enhanced patches to further improve performance.

4.1. Additional Discussion Based on Prior Work.

Our findings are in line with recent literature. Truong et al. [3] evaluated multiple detectors (Faster-RCNN, Cascade-RCNN, etc.) on the VisDrone dataset and found that SR-enhanced inputs (via RCAN) only partially improved performance compared to HR images. Notably, detection on SR images still underperformed HR baselines by a margin of 3–6% in AP metrics, implying that SR alone cannot fully recover HR-level features.

Shermeyer et al. [2] also reported that SR preprocessing often alters feature distributions and edge characteristics, potentially harming detection in satellite imagery. They concluded that SR “often fails to meaningfully improve detection,” and recommended task-specific SR or hybrid fusion strategies.

In underwater imagery, Nguyen et al. [1] proposed simultaneous enhancement and SR, but emphasized that traditional SR (e.g., for perceptual quality) may not always

help object detection. They advocated for joint training pipelines or classifier-aware SR methods.

Together, these studies support our key insight: SR must be carefully integrated into detection pipelines, with consideration for domain shift and label reliability. Blindly applying SR before detection—without retraining or adaptation—may degrade performance despite visual improvements.

References

- [1] Phong Nguyen, Le Thanh Tran, Son Lam Nguyen, et al. Simultaneous enhancement and super-resolution of underwater imagery for improved visual perception. *Sensors*, 22(2):421, 2022. [4](#)
- [2] Jacob Shermeyer and Adam Van Etten. The effects of super-resolution on object detection performance in satellite imagery. In *CVPR Workshops*, 2019. [4](#)
- [3] Ngan T Truong, Nguyen D Vo, and Khang Nguyen. The effects of super-resolution on object detection performance in an aerial image. In *2020 7th NAFOSTED Conference on Information and Computer Science (NICS)*, pages 256–260. IEEE, 2020. [4](#)